

VARSHINI SHERI

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SUMMARY

Software engineer with hands-on experience building production-grade agentic AI systems, RAG pipelines, and cloud-native infrastructure on Azure and AWS, with applied exposure to cybersecurity automation and enterprise risk analytics.

EDUCATION

Master of Science in Computer Science, University of Illinois at Chicago (*GPA: 3.8/4*) May 2026
Bachelors in Computer Science and Engineering, Jawaharlal Nehru Technological University (*GPA: 8.8/10*) May 2024

EXPERIENCE

Software Developer UIC College of Medicine, Chicago, Illinois, USA July 2025 - Present
Tech Stack: JavaScript, React, REST APIs, MySQL

- Built modular React components and UI modules across 20+ production websites, making layout decisions for new pages and revising existing pages for visual consistency, with attention to accessibility (WCAG).
- Automated institutional data processing and reporting workflows using shell scripting, reducing manual reporting time by 60%.

AI Engineer G19 Studios, Chicago, Illinois, USA January - December 2025
Tech Stack: Python, React, REST APIs, Azure Functions, Cosmos DB, SQL, GPT4, LangChain, LangGraph

- Architected a multi-agent NL-to-SQL system with LangGraph, orchestrating query execution over operational databases and reducing report generation time by 80%.
- Built a RAG pipeline over conversational history using CosmosDB for session memory retrieval, improving response accuracy by 45% for analytical queries.
- Implemented async Azure Functions orchestration with retry logic, governance checks, and query safety validation to evaluate and constrain multi-step execution reliably at scale.
- Integrated the agent pipeline into CI/CD workflows, including integration tests validating API contracts and end-to-end execution accuracy, reducing manual release effort and deployment errors.

Software Development Engineer MSCI, Pune, India January – June 2024
Tech Stack: Python, Azure Functions, Azure DevOps, Akamai WAF, Layer7, Qualys, Wiz

- Designed and deployed event-driven Azure Functions in Python to automate multi-region Akamai WAF policy updates, reducing manual SecOps effort by 50%. Deployed via Azure DevOps CI/CD.
- Implemented RESTful integration of ServiceNow with Layer7 workflows, reducing MTTA for critical security incidents by 30%. Supported end-to-end incident response.
- Performed vulnerability management for on-prem and cloud environments(Azure, AWS, GCP); built analysis pipelines over scan outputs to track risk posture and maintain SLA compliance across 6 teams, supported remediation.
- Prototyped an LLM pipeline over historical incident logs to predict emerging threat models for SecOps automation.

PROJECTS

RAG Cloud – Distributed Semantic Search System | *Scala, Apache Flink, Spark, AWS, Neo4j, Lucene, CloudWatch*

- Built a distributed RAG-based search system for research documents, implementing batch and streaming ETL pipelines using Spark and Flink to index large-scale document data, supporting both offline processing and incremental updates.
- Built an LLM based evaluation pipeline to benchmark GraphRAG against vector-only retrieval, validating a 25% improvement in semantic query relevance.
- Deployed Spark batch workloads on AWS EMR and Flink/GraphRAG services on EKS(Kubernetes), with structured logging and CloudWatch monitoring for scalable production.

Secure Distributed Object Storage System | *Python, FastAPI, Linux, Docker, Prometheus, AWS*

- Designed a Byzantine fault-tolerant distributed object store using fingerprint verification, quorum voting, and erasure coding; deployed across EC2 instances via FastAPI with Prometheus for real-time node monitoring.

Conversational LLM Fine-Tuning | *Python, PyTorch, Hugging Face, LoRA, CUDA*

- Fine-tuned Gemma-3 4B using Unsloth, QLoRA (4-bit quantization), and LoRA adapters on a 100K conversational dataset, enabling efficient single-T4 GPU training.
- Implemented response-only training by masking instruction tokens during loss computation, improving fine-tuning accuracy by 83%.

TECHNICAL SKILLS

Programming & Scripting: Python, C/C++, C#, Bash, Linux, Java, Spring Boot, SQL, JavaScript, React, TypeScript

Security & Networking: OAuth 2.0, WAF, Akamai, Nmap, Wireshark, Qualys, Wiz, TCP/IP

Backend & APIs: Django, Flask, RESTful APIs, GraphQL, Azure Functions, Postman

AI & ML: Scikit-learn, PyTorch, TensorFlow, Hugging Face, PEFT, LangChain, LangGraph, Neo4j, CUDA

Cloud & DevOps: Azure, AWS (EC2, S3, EKS, EMR), Docker, Kubernetes, Prometheus, CI/CD Pipelines

Data & Tools: Apache Spark, Flink, Lucene, Hadoop, Snowflake, PostgreSQL, MySQL, MongoDB, Git